1. (a)

we say a vector space W is a quotient of V modulo U.

if there is an onto map $V \xrightarrow{q} W \to 0$ s.t.

$$0 \to U \xrightarrow{i} V \xrightarrow{q} W \to 0 \text{ is s.e.s.}$$

such W is called quotient space of V modulo U, denoted $W = V/U$.

(b) $V/U = \{ v+U \mid v \in V \}$.

$$q : V \to V/U \qquad v \mapsto v+U$$

is onto and $\ker(q) = U$.

First prove q is onto. take $\forall \alpha = v+U \in V/U \Rightarrow q(v) = v+U$.

Then we prove $\ker(q) = U$.

Lemma: $v+U = U \Leftrightarrow v \in U$.

proof:

"$\Rightarrow$" $v = v+0 \in v+U$. Since $v+U = U \Rightarrow v \in U$.

"$\Leftarrow$" Since $v \in U \Rightarrow$ take $\forall v' \in U$ $v+v' \in U$. $\Rightarrow v+U \subseteq U$.

take $\forall v' \in U \Rightarrow v - v' \in U \Rightarrow$ we can always find $v'' \in U$ s.t. $v+v'' = v'$

$\Rightarrow U \subseteq v+U \Rightarrow v+U = U$.

$\Rightarrow q(v) = U \Leftrightarrow v \in U \Rightarrow \ker(q) = U$.

(c) The natural induced map: $\overline{T} : V/\ker(T) \to \text{image}(T)$

$$v + \ker(T) \mapsto T(v).$$

First we prove it is well-defined. take $\forall v+\ker(T) = v'+\ker(T)$

we want to prove $T(v) = T(v')$. Since $v - v' \in \ker(T) \Rightarrow T(v-v') = 0$

$\Rightarrow T(v) - T(v') = 0 \Rightarrow T(v) = T(v')$.

Then prove $\bar{T}$ is linear.

① Take $\forall\, v+\ker(T),\ u+\ker(T)$

$\bar{T}(v+\ker(T)+u+\ker(T)) = \bar{T}(v+u+\ker(T)) = T(v+u) = T(v)+T(u) = \bar{T}(v+\ker(T)) + \bar{T}(u+\ker(T))$

② Take $\forall \lambda \in \mathbb{F}$. $v+\ker(T)$ where $\lambda \neq 0$

$\bar{T}(\lambda(v+\ker(T))) = \bar{T}(\lambda v+\ker(T)) = T(\lambda v) = \lambda T(v) = \lambda \bar{T}(v+\ker(T))$

Then prove $\bar{T}$ is injective:

if $\bar{T}(v+\ker(T)) = \bar{T}(u+\ker(T))$, then $\bar{T}(v+\ker(T)) = T(v) = \bar{T}(u+\ker(T)) = T(u)$

$\Rightarrow v-u \in \ker(T) \Rightarrow v+\ker(T) = u+\ker(T)$.

Then prove $\bar{T}$ is surjective:

Take $\forall\, T(v)$. we have $v+\ker(T)$ s.t. $\bar{T}(v+\ker(T)) = T(v)$

Therefore $\bar{T}$ is isomorphism.

2.

(a) $\mathcal{B}$ is a basis for V if:

① $\text{span}(\mathcal{B}) = V$

② $\mathcal{B}$ is linearly independent.

(b) If V is finite dimensional. then basis exists and

finite with same number of elements like $\dim(V)$

(c) $\mathcal{B}_U$ is a basis for U. denoted as $\{b_1, \cdots, b_m\}$.

Then the basis $\mathcal{B}$ of V is $\{b_1, \cdots, b_m, c_1, \cdots, c_n\}$ where $C = \mathcal{B} \setminus \mathcal{B}_U = \{c_1, \cdots, c_n\}$

$\Rightarrow \mathcal{B}_{V/U} = \{q(v) \mid v \in C\} = \{v + U \mid v \in C\} = \{c_1 + U, \cdots, c_n + U\}$

First improve $\mathcal{B}_{V/U}$ spans V/U.

Take $\forall x = v + U \in V/U$. $\Rightarrow x = r_1 b_1 + \cdots + r_m b_m + s_1 c_1 + \cdots + s_n c_n + U$.

Since $r_1 b_1 + \cdots + r_m b_m \in U \Rightarrow x = s_1 c_1 + \cdots + s_n c_n + U = s_1(c_1 + U) + \cdots + s_n(c_n + U)$

$\Rightarrow \forall x \in V/U$ can be represented by linear combination of $\mathcal{B}_{V/U}$.

$\Rightarrow \mathcal{B}_{V/U}$ is a basis of V/U.

Next to prove $\mathcal{B}_{V/U}$ is linear independence.

Because $\mathcal{B}$ is a basis $\Rightarrow b_1, \cdots, b_m, c_1, \cdots, c_n$ linear independence $\Rightarrow c_1, \cdots, c_n$ linear independence

if $w_1(c_1 + U) + \cdots + w_n(c_n + U) = 0 + U \Leftrightarrow w_1 c_1 + \cdots + w_n c_n + U = U$

$\Leftrightarrow w_1 c_1 + \cdots + w_n c_n = 0$. Since $c_1, \cdots, c_n$ linear independence $\Rightarrow c_1 + U, \cdots, c_n + U$ linear independence.

3.

(a) Def of Dual Space: $V^* = \mathrm{Hom}_{\mathbb{F}}(V, \mathbb{F})$.

(b)

First we prove B^* spans V^*. Assume $B = \{b_1, \cdots, b_n\}$ a basis of V.

$B^* = \{\varphi_{b_1}, \cdots, \varphi_{b_n}\}$

Take $\forall \varphi \in V^*$. $\varphi(u) = w$

$u = a_1 b_1 + \cdots + a_n b_n$

$\Rightarrow \varphi(u) = a_1 \varphi(b_1) + \cdots + a_n \varphi(b_n)$

Since $\varphi_{b_i}(u) = \varphi_{b_i}(a_1 b_1 + \cdots + a_n b_n) = a_1 \varphi_{b_i}(b_1) + \cdots + a_n \varphi_{b_n}(b_n) = a_i$.

$\Rightarrow \varphi(u) = \sum_{i=1}^{n} \varphi_{b_i}(u) \varphi(b_i)$ Therefore B^* spans V^*.

Next we prove B^* linear independence.

Consider $\sum_{i=1}^{n} a_i \varphi_{b_i}(u) = 0$ for $\forall u$. $u = \sum_{i=1}^{n} c_i b_i$.

$\Rightarrow \sum_{i=1}^{n} a_i \varphi_{b_i}(u) = \sum_{i=1}^{n} a_i \varphi_{b_i}(\sum_{j=1}^{n} c_j b_j) = \sum_{i=1}^{n} a_i c_i = 0$ for $\forall c_i$.

Let $c_i = 1$, other $c_{j \neq i} = 0 \Rightarrow a_i = 0$, $i = 1, 2, \cdots, n \Rightarrow a_1 = \cdots = a_n = 0$

$\Rightarrow B^*$ is a basis of V^*.

(c) $\psi : \mathbb{F}^n_{row} \longrightarrow (\mathbb{F}^n_{col})^*$

$(a_1, \cdots, a_n) \mapsto \left((x_1, \cdots, x_n)^T \mapsto \sum_{i=1}^{n} a_i x_i \right)$

First we prove ψ is well-defined. $u = (a_1, \cdots, a_n)$

$\psi(u) = \varphi$ where φ is a map $\varphi((x_1, \cdots, x_n)^T) = \sum_{i=1}^{n} a_i x_i$

$\psi(u)(x+y) = \varphi(x+y) = \sum_{i=1}^{n} a_i (x_i + y_i) = \sum_{i=1}^{n} a_i x_i + \sum_{i=1}^{n} a_i y_i = \varphi(x) + \varphi(y)$.

$\Rightarrow \psi(u)$ is linear.

Then prove ψ is linear.

$\psi(u_1 + u_2)(x) = \sum_{i=1}^{n} (a_i + b_i) x_i = \sum_{i=1}^{n} a_i x_i + \sum_{i=1}^{n} b_i x_i = \psi(u_1)(x) + \psi(u_2)(x)$ for $\forall x \Rightarrow \psi(u_1 + u_2) = \psi(u_1) + \psi(u_2)$

$\psi(\lambda u_1)(x) = \sum_{i=1}^{n} \lambda a_i x_i = \lambda \sum_{i=1}^{n} a_i x_i = \lambda \psi(u_1)(x) \Rightarrow \psi(\lambda u_1) = \lambda \psi(u_1)$

Then prove L is surjective

Take map $\varphi \in (\mathbb{F}^n_{col})^*$. denoted as $(x_1, \cdots, x_n)^T \mapsto v$, where v can always be written as $v = \sum_{i=1}^{n} a_i x_i$ for some $(a_1, \cdots, a_n)$

if want $L(r) = \varphi \Rightarrow r = (a_1, \cdots, a_n) \Rightarrow L(r) = \varphi$.

Then prove L is injective want to prove $\ker(L) = \{0\}$.

If $L(r)(x) = 0$ for $\forall x$. $L(r)(x) = 0 \Leftrightarrow r = (a_1, \cdots, a_n) = (0, 0 \cdots, 0)$

otherwise if $a_i \neq 0$. then let $x_1 = \cdots = x_{i-1} = x_{i+1} = \cdots = x_n = 0$, $x_i = 1$. $\Rightarrow L(r)(x) \neq 0$.

$\Rightarrow \ker(L) = \{0\}$

Therefore such L :

$$L : \mathbb{F}^n_{row} \longrightarrow (\mathbb{F}^n_{col})^*$$
$$(a_1, \cdots, a_n) \mapsto \left((x_1, \cdots, x_n)^T \mapsto \sum_{i=1}^{n} a_i x_i \right)$$

is a natural isomorphism.

4.

(a) Multilinearity:

$$\mathcal{A}(\lambda v+\mu w, u)=\lambda\mathcal{A}(v,u)+\mu\mathcal{A}(w,u).$$

$$\mathcal{A}(u,\lambda v+\mu w)=\lambda\mathcal{A}(u,v)+\mu\mathcal{A}(u,w)$$

Skew-Symmetry:

$$\mathcal{A}(v,w)=-\mathcal{A}(w,v)$$

(b)

(1) Let $e_1=(1,0)$ $e_2=(0,1)$.

$v=(a,b)$, $u=(c,d)$ $\Rightarrow v=ae_1+be_2$, $u=ce_1+de_2$.

$$\mathcal{A}(v,u)=\mathcal{A}(ae_1+be_2, ce_1+de_2)=ac\mathcal{A}(e_1,e_1+de_2)+b\mathcal{A}(e_2,ce_1+de_2)$$

$$=ac\mathcal{A}(e_1,e_1)+ad\mathcal{A}(e_1,e_2)+bc\mathcal{A}(e_2,e_1)+bd\mathcal{A}(e_2,e_2)$$

Since $\mathcal{A}(x,x)=-\mathcal{A}(x,x)\Rightarrow\mathcal{A}(x,x)=0$.

$\Rightarrow\mathcal{A}(v,u)=ad\mathcal{A}(e_1,e_2)-bc\mathcal{A}(e_1,e_2)=(ad-bc)\mathcal{A}((1,0),(0,1))$

$\Rightarrow$ It is completely determined by the value $\mathcal{A}((1,0),(0,1))$.

(2) Let $D:\mathbb{F}_{row}^2\times\mathbb{F}_{row}^2\to\mathbb{F}$

$$D((a,b),(c,d))=ad-bc.$$

$\Rightarrow\forall\mathcal{A}(v,u)=D(v,u)\mathcal{A}((1,0),(0,1))$ where $v,u\in\mathbb{F}_{row}^2$.

$\Rightarrow\forall\mathcal{A}\in\Lambda(\mathbb{F}^2)=D\mathcal{A}((1,0),(0,1))\Rightarrow D$ spans $\Lambda(\mathbb{F}^2)$.

$\Rightarrow\{D\}$ is a basis $\Rightarrow\dim(\Lambda(\mathbb{F}^2))=1$

(3) $\mathcal{A}_1((1,0),(0,1))=1\Rightarrow\mathcal{A}_1((x,y),(x',y'))=(xy'-yx')\mathcal{A}_1((1,0),(0,1))=xy'-yx'$

(c) $M[\mathcal{A}]((1,0),(0,1))=\mathcal{A}((1,0)\begin{pmatrix}a&b\\c&d\end{pmatrix},(0,1)\begin{pmatrix}a&b\\c&d\end{pmatrix})=\mathcal{A}_1((a,b),(c,d))$

$=(ad-bc)\mathcal{A}_1((1,0),(0,1))=ad-bc=d(M)\cdot\mathcal{A}_1((1,0),(0,1))\Rightarrow d(M)=ad-bc.$

Therefore: $d(M)=ad-bc$.